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REVIEW ARTICLE

On terrain traversability analysis in unstructured environments: recent advances in forest applications

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Abstract
In the field of mobile robotics and autonomous systems, the ability to navigate in unstructured environments is a major challenge. Unpredictable landscapes consisting of irregular and possibly unconsolidated terrain, especially in natural environments such as forests, require comprehensive analysis and innovative solutions for robot perception and control. This article provides an introduction to the field of traversability analysis, with a particular focus on recent advances tailored to forest environments. It presents both the classical and earlier techniques as well as current state-of-the-art methods that enable the adaptability and efficiency of robots in maneuvering through complex, unstructured landscapes. By highlighting and discussing relevant methods, this paper aims to succinctly provide the reader with fundamental knowledge regarding traversability analysis in a forestry context. Finally, a roadmap for the future use of traversability analysis methods in forestry environments is presented.

Keywords Traversability analysis · Unmanned ground vehicles · Artificial perception · Forest

1 Introduction

Traversability analysis in the context of field robotics refers to the process of assessing whether a given terrain or environment is navigable for a robot. It holds immense significance in the domain of autonomous navigation, particularly in unstructured landscapes.

This short and focused review paper addresses the intricacies of terrain traversability analysis, delving into recent advances developed specifically for forest environments. By examining both classical and modern techniques, the aim is to provide the reader with a general understanding of the field. From computer vision-based approaches using neural networks to LiDAR-based methods and multisensor fusion strategies, various methods that are driving progress in terrain traversability analysis are explored. In addition, recent breakthroughs are highlighted, including adaptive traversability concepts and meta-learning frameworks that promise to reshape the landscape of robotic navigation in complex and unstructured terrain. Through a concise yet detailed analysis, this paper aims to provide an overview of the current state of the art in traversability analysis and lay the foundation for future innovations in autonomous robotics.

Forest environments are some of the most difficult terrains to navigate. Among many other nuances, the consistency of the ground is difficult to assess and is not constant in space. To make matters worse, meteorological conditions can abruptly change the properties of the soil, which also affects its stability over time. In addition, dense vegetation can occlude sensors and hide potentially dangerous obstacles such as large rocks or deep holes. This particular set of challenges is found in almost no other terrain classes (orchards, structured forests, roads, sidewalks, etc.). Orchards, for instance, may have unstructured terrains, but their tree arrangement is typically well defined—when properly maintained—which significantly simplifies the navigation task. This inherent complexity of forest terrain highlights the importance of robust traversability analysis, which serves as a critical perception layer. By providing accurate traversability information to planning and decision-making systems, autonomous systems can achieve significantly improved robustness and adaptability. This approach not only addresses the unique challenges posed by forest environments, but also extends

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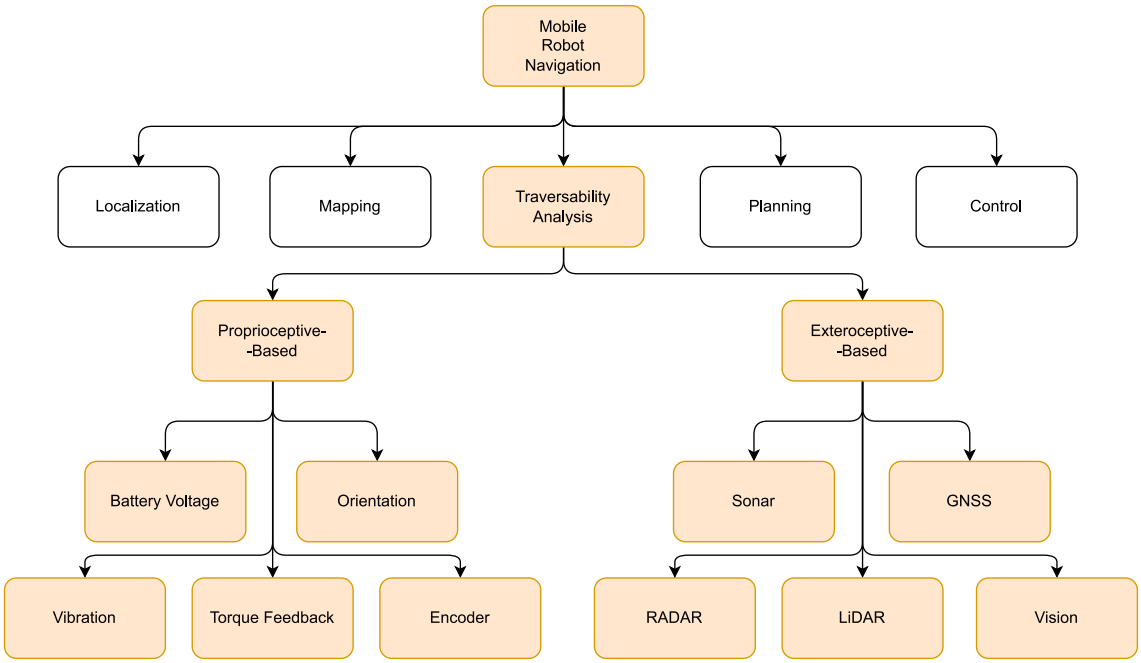


Fig. 1 Typical mobile robot navigation taxonomy encountered in the studied works with a special focus on traversability analysis

to other domains such as search-and-rescue operations and autonomous monitoring.

Integrating traversability analysis into practical systems is key for robust real-world applications. For example, in search-and-rescue scenarios, effective traversability analysis allows autonomous agents to adapt to unstructured landscapes while prioritizing safety and efficiency. Similarly, in environmental monitoring, these methods enable consistent operation under dynamic conditions. These principles are also crucial for extraterrestrial missions, where terrain complexity is coupled with additional constraints. Addressing how traversability analysis integrates with perception and decision-making is essential to deploy real-world systems capable of reliably operating in such demanding environments [1].

To clearly delineate the scope of this survey, the focus is primarily on ground robots and systems with potential applications in forest environments. Figure 1 represents a taxonomy of traversability analysis methods which encapsulates the typical high-level approaches found in the examined articles. For this survey, works that showed little innovation or did not deal specifically with traversability in unstructured environments were excluded.

The article starts with a review of foundational concepts and methods and progresses to examine recent advances and emerging trends. Subsequent sections focus on particularly relevant techniques and breakthroughs, illustrating their application and effectiveness in forest environments. The final sections discuss the implications of these advances for future research and development in autonomous robotics, providing a concise yet comprehensive guide to the current landscape and potential future directions for terrain traversability analysis.

2 Early terrain classification and traversability approaches

Borges et al. [2] define traversability analysis as "a multidisciplinary field combining robotics with image and signal processing, feature extraction, machine learning, three-dimensional (3D) mapping, and 3D geometry."

Papadakis [3] defines it as "the capability of a ground vehicle to reside over a terrain region under an admissible state wherein it is capable of entering given its current state, this capability being quantified by taking into account a terrain model, the robotic vehicle model, the kinematic constraints of the vehicle, and a set of criteria based on which the optimality of an admissible state can be assessed."

Inspired by these definitions and the current state of this field of research, in this article we will consider traversability analysis as **the ability of a mobile robot to compute the feasibility and/or effort associated with the traversal of a certain portion of terrain.**

Since the twentieth century, traversability analysis has been used as an integral part of most autonomous navigation modules [4, 5], particularly in unstructured environments such as forests, worksites [6] and extra-planetary terrains [1, 7, 8].

Gennery [4] introduces a geometric method for calculating estimates of smoothed and interpolated height, slope, and roughness from 3D data, such as those obtained through stereo vision or LiDAR systems, to aid vehicle path planning. The method, which incorporates error propagation to estimate accuracy, produces results at uniformly distributed points that are advantageous for determining traversable areas. Also discussed is the iterative nature of the calculations, the potential use of these calculations in developing a cost function for path planning, and the contribution of these methods to optimizing vehicle navigation in different terrains. Tests were performed using an HMMWV (an instrumented army Humvee vehicle).

Howard et al. [5] discuss the use of fuzzy logic for on-board terrain assessment to infer terrain characteristics, enabling robots to navigate without prior knowledge of the environment, ultimately aiming to enhance the navigation strategies of autonomous mobile robots. The proposed approach was tested on a Pioneer AT mobile rover equipped with stereo cameras. Results showed that the technique enables robots to navigate terrains with no a priori knowledge of the environment.

Early traversability analysis methods were developed mainly for autonomous planetary exploration, comprising the at-the-time state of the art for navigating difficult terrain, such as those found on extraterrestrial bodies. For example, in an initial attempt to improve the computational capabilities of Mars rovers, Goldberg et al. [1] proposed a shift from radiation-hardened to commercial processors. This approach was intended to address the limitations of available technology in terms of the amount of scientific exploration possible within a limited time frame, emphasizing the need for more efficient processing in autonomous stereo vision systems.

Additionally, the adoption of soft computing techniques to assess terrain traversability marked an early attempt into leveraging visual cues and imaging analysis for navigation, as shown by Shirkhodaie et al. [7]. Early classifiers, employing rule-based, neural network, and fuzzy logic approaches, aimed to segment natural terrains into navigable sub-regions, providing a rudimentary yet crucial framework for path planning in unstructured environments.

Wang et al. [8] proposed the development of a stereo-based mapping and analysis framework, incorporating algorithms for disparity expansion and digital elevation mapping, reflecting an initial step toward enhancing the navigational precision and safety of microplanetary rovers on rough terrains. These early efforts are a good indicator of the challenges of robust terrain traversability analysis and path planning with limited sensor data and computational resources. Collectively, these and many other methodologies have set in motion the development of autonomous navigation strategies, from basic computational adjustments and terrain classification to more comprehensive mapping and analysis frameworks.

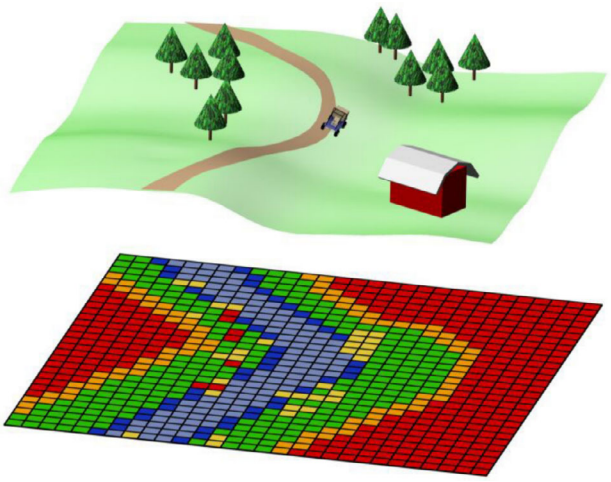


Fig. 2 Traversability grid utilized in Crane et al. [9] work. Image extracted from the original publication

Over the years, several approaches for terrain classification have been proposed, with earlier ones mostly relying on techniques based on optical sensors, two-dimensional representations such as traversability grids [9] (see Fig. 2) and others supported by fuzzy logic [10–12]. The integration of local and global traversability indices has marked an important step in improving the assessment of terrain safety, as shown in [10]. Local indices, derived from on-board sensors, evaluated immediate obstacles and surface conditions, while global indices, derived from topographic maps, considered long-range terrain features such as hills and lakes. These indices, which were classified into fuzzy sets ranging from poor to high traversability, served as the basis for both sensor-based and map-based navigation behaviors to avoid hazards.

Stereo vision-based estimation techniques used sloped planes or digital elevation models to map surrounding terrain, adjusting the methodology based on data availability. This dual approach aimed to increase the range and detail of the terrain map and marked early attempts to balance between sensor input and computational modeling for safer robot navigation [12].

Learning-based methods have also been proposed, ranging from real-time terrain characterization using image data and fuzzy logic to quantify traversability indices [13] to segmentation of LiDAR data to identify vegetation, obstacles, and ground surfaces using local three-dimensional point cloud statistical measures [14]. Other methods used a single compact camera and used a bag of visual words (BOVW) approach with support vector machine (SVM) classifiers to demonstrate adaptability to changes in illumination and terrain complexity [15]. Other approaches have also used

proprioceptive data, such as motor feedback and platform vibrations, to estimate the characteristics of the navigated environment. For example, Bermudez et al. [16] presented an algorithm for terrain classification that uses vibration data from the on-board IMU and motor control data from back-eletromotive force (EMF) sensing and magnetic encoders. The proposed technique demonstrated sufficient efficiency to operate in real time on a mobile robot's microcontroller and exhibited remarkable accuracy in terrain classification.

Other research directions have also been explored that use classical obstacle avoidance algorithms such as the vector field histogram (VFH) [10] in traversability analysis [17, 18].

3 Terrain classification and traversability cost estimation

In the field of autonomous robots, accurate terrain classification involves the categorization of different terrain types based on their physical properties and characteristics, playing a crucial role in enabling robot navigation and effective interaction with the environment.

Exteroceptive data collected by sensors such as cameras and depth sensors provide rich information that can be used to classify the terrain, as shown in Fig. 3. Computer vision techniques, including image processing, feature extraction, and machine learning algorithms, are commonly employed to analyze visual data and classify terrain based on color, texture, shape, or other visual cues [19, 20].

Prágr et al. [22] proposed a method that uses aerial imagery above the canopy (in areas with little to high vegetation/trees) to create a terrain model, which is then used to extract terrain features where ground units will navigate. Based on previous traversal experience of ground units in similar environments, the traversal cost model that was learned is used to predict the traversal cost of the new environment and plan an efficient path connecting all points of interest. The method achieved promising results in multiple scenarios, where the cost prediction results were visually coherent and the paths mainly avoid higher cost terrains whenever possible. As future work, the authors aim to extend the approach to use a real-time communication mechanism between the ground and aerial vehicles so that it can learn online.

Proprioceptive data collected by sensors such as IMUs are also commonly used for terrain classification [23–25]. For example, Faigl et al. [24] use self-organizing maps in conjunction with unsupervised learning to estimate the cost of terrain traversal by a hexapod walking robot. Using proprioceptive signals such as energy consumption and stability of the robot in combination with exteroceptive information captured by an RGB-D camera, the authors succeed in predicting the cost of traversing terrain that has been seen but not yet visited. Using proprioceptive signals, such as energy consumption and robot stability, combined with exteroceptive information collected by an RGB-D camera, the authors managed to predict the traversal cost of seen but not yet visited terrain.

Waibel et al. [25] presented a comparison between two different approaches to estimate the roughness of the terrain and encoding this information in continuous costmaps: one is based solely on LiDAR and calculates terrain cost directly from point clouds using a statistical approach; the other is a learning-based approach that predicts an IMU output from geometric point cloud data using a 2D Convolutional Long Short-Term Memory (LSTM) neural network. The results indicate that both approaches demonstrate versatility for local and global planning, with the learning-based approach not heavily relying on specific terrain features and avoiding the need for threshold tuning on the costmap, thus enhancing its generality. In most cases, the learning-based approach either matches or surpasses the performance of other approaches. However, both methods achieve similar results and significantly outperform methods based on a typical binary costmap by a considerable margin.

Eder et al. [26] propose a pipeline for traversability analysis and cost estimation for a robotic system on different terrain classes using locomotion data, multidimensional terrain classes, and segmentation of the environment to calculate robot-dependent traversability costs. Although experiments have shown that the results are dependent on the quality and

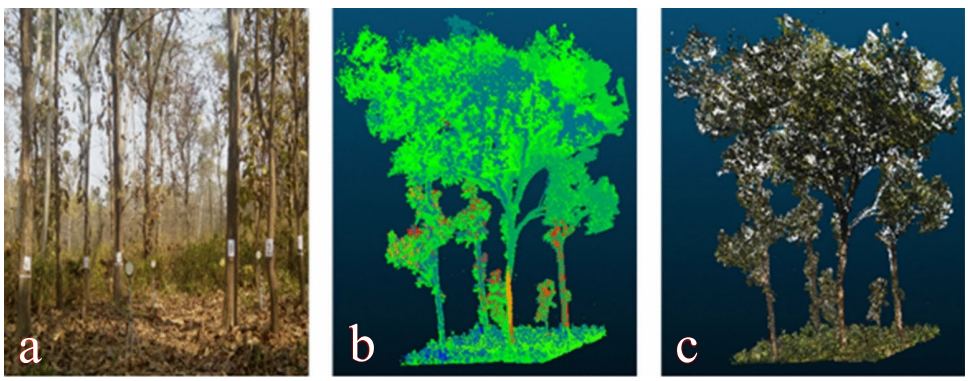


Fig. 3 An RGB image of a forest scenario on the left (**a**), the respective point cloud created by a LiDAR sensor in the middle (**b**) and the depth model extracted by an RGB-D camera in the right (**c**). Image extracted from Bhandari et al. [21]

quantity of the recorded locomotion data, the computed costs reflect the capabilities of the robotic systems well, thus confirming the usefulness of the proposed pipeline.

Energy consumption and the ability to reliably estimate it is also relevant, enabling more nuanced metrics to be considered for traversability analysis, such as the estimated energy consumption for a certain path to be executed. On this front, many works indirectly estimate energy within their traversability analysis pipelines. However, Wei et al. [27] proposed a learning-based function that takes terrain geometry and robot motion direction as input, yielding the expected energy consumption. This method achieved a prediction error of 12% compared to the ground truth, outperforming a physics-based baseline method from the literature by 10%, which struggles to predict energy consumption in uneven terrains.

Svenson et al. [28] also explored the concept of energy consumption in heavy-duty logging trucks in unstructured environments and presented a function that describes the impact of gradient and road surface roughness on fuel consumption. Furthermore, they found and showed that 77% of the variation in fuel consumption could be explained by these two variables. Taking into account this relationship, Carvalho et al. [29] presented a method that indirectly estimates energy consumption and attempts to reduce it by minimizing the terrain gradient experienced by the robot during the execution of a chosen path.

4 Latest advances and trends

The way researchers approach this problem and the techniques they employ can be fundamentally different from method to method. For that reason, we currently find promising results in traversability analysis generated by methods based on computer vision techniques. These typically use neural networks in order to semantically segment images and decide which paths are traversable [30–32]. This kind of techniques tends to require a large quantity of annotated images to train neural networks, which is usually hard to acquire considering that the labeling process is usually performed by hand. However, there are notable exceptions, such as self-training networks [33], in which the authors propose a method for traversability analysis based on a self-training algorithm that eliminates the hand labeling process. In the aforementioned work, the authors show promising results in structured urban scenarios in which their technique manages to outperform supervised learning-based methods.

Also, methods based on Time of Flight (ToF) sensors such as LiDAR [34–37] using various strategies such as normal distributions transform analysis [36, 37] or navigation meshes [38, 39] have been used with success.

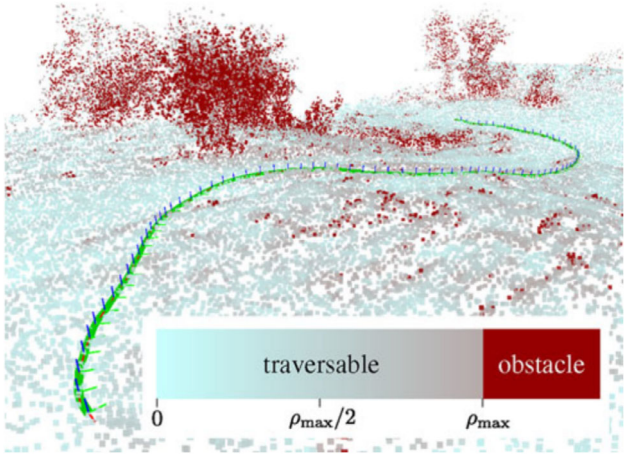


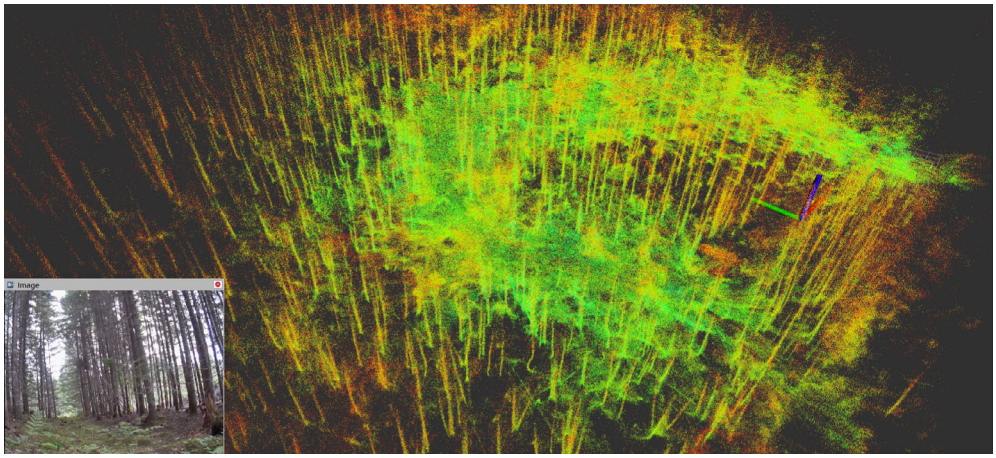
Fig. 4 Output of the approach proposed by Krusi et al. [40]. Pictured reproduced from the mentioned work

A very promising LiDAR-only work has been presented by Krusi et al. [40], in which the authors assess the traversability of the terrain directly from an unordered point cloud (see Fig. 4). This is of great importance, as it eliminates the need for explicit modeling of the terrain with a specific framework and avoids all associated computational costs. Furthermore, they can also perform 6D motion control and path planning in complex unstructured and dynamic scenarios. The proposed approach was tested in indoor and outdoor scenarios, involving structured and unstructured terrain. Results show that the robot can autonomously navigate the environment, covering 6D waypoints with minimal human intervention, often traveling upwards of 1000 m, demonstrating the robustness of the approach.

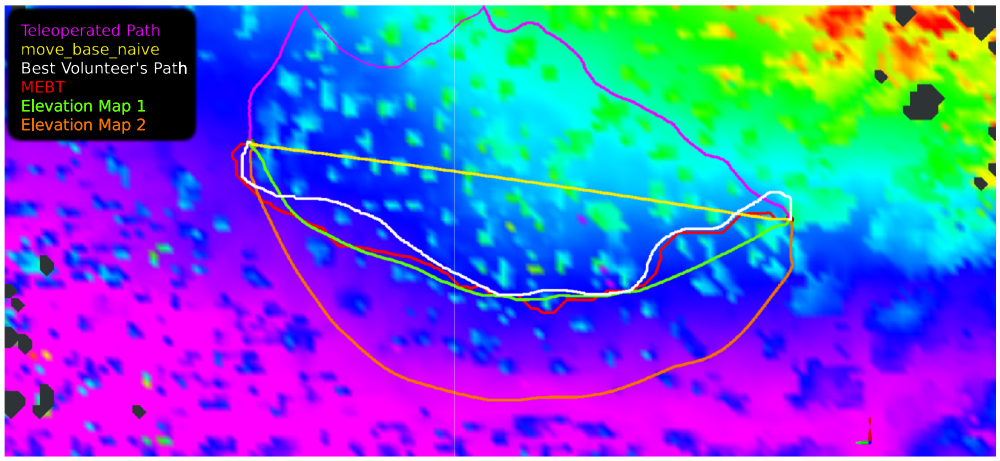
Another LiDAR-only geometric approach is published by Carvalho et al. [29], in which the authors proposed a new method for analyzing traversability and planning paths using 3D point clouds to detect obstacles and assess terrain gradients. It devises routes that avoid steep slopes, improving fuel efficiency and equipment longevity, showcasing superior performance in both simulations and real-world forest tests against other approaches. The generated paths closely match those chosen by human experts in real-world experiments performed using the Montmorency dataset[1] [41] (see Fig. 5b), proving the practical applicability of the method.

There is also the case where multiple sensor modalities are used simultaneously to achieve higher robustness and accuracy in their traversability estimations, as is the case in [42], in which the authors propose an algorithm for efficiently detecting obstacles by cooperatively fusing stereo vision and laser scan data to filter out false positives from ground hits, enhancing autonomous vehicle navigation in real-world industrial environments. Wang et al. [43] fuse the measurements from

[1] https://norlab.ulaval.ca/research/montmorencydataset.



(a) Visualization of the environment in the Montmorency dataset as a point cloud.



(b) One of the main results reported in Carvalho et al. [29], showing how closely the path generated based on the MEBT traversability analysis (the method proposed in the article, in red) resembles the one generated by the best expert user according to the defined metrics (in white).

Fig. 5 Mechanical effort based traversability (MEBT) method presented in [29] and the main results obtained in the study

a 2D LiDAR and an RGB-D camera to develop a hierarchical path planning system for autonomous robotic exploration of uneven indoor spaces. They demonstrate robust and safe navigation while avoiding stairs, which could be used in wheelchair navigation, for example.

4.1 NDT-based approaches

Normal Distributions Transform (NDT) allows for precise fusion of multiple sensor data and includes a probabilistic representation of one or more characteristics of the environment (occupancy, traversability, risk, etc.) and its uncertainty [44]. As such, NDT-based methodologies have been having great success in estimating terrain traversability in rough foretry terrain.

Ruetz et al. [34] addressed the ongoing challenge of autonomous navigation in dense vegetation, presenting a novel Forest Traversability Map (FTM) method that extends the state-of-the-art NDT-TM method of Ahtiainen et al. [36]. FTM essentially differentiates itself by employing a 3D voxel representation, a robust color fusion method to handle occlusion and lighting changes, and the integration of multi-return LiDAR measurements. Also, the authors introduce and evaluate features such as a 3D probabilistic map representation, adjacency considerations, and an experimental validation on challenging vegetated environments. As such, the proposed method is compared against and outperforms the NDT-TM [36] state-of-the-art method, demonstrating a remarkable improvement of 20 percentage points in F1 score on real-world data. An example of an FTM, as well as a comparison with NDT-TM, is found in Fig. 6.

Ahtiainen et al. [36] presented a method that incrementally expands Normal Distributions Transform-Occupancy Map (NDT-OM) maps by augmenting their geometric information with statistical data of the permeability and reflectivity of each cell. This is an innovative way of tackling rough terrain

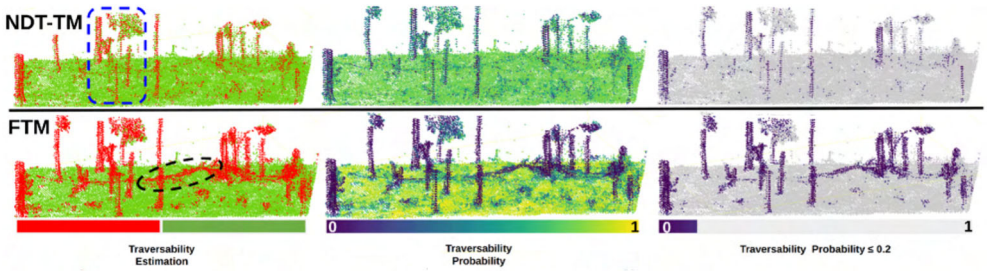


Fig. 6 A comparison between the outputs of NDT-TM (top row) and NDT-FTM (bottom row). Image adapted from [34]

navigation through soft vegetation and leads to promising results, showing that sparsely vegetated areas are classified as traversable terrain, where otherwise would be partially or fully non-traversable, as seen in Fig. 7.

Another seminal recent work that employs NDT as part of a mapping strategy is presented by Xue et al. [45], proposing a technique that also applies Bayesian Generalized Kernel Inference, as in [35]. Their work presents a LiDAR-based traversability analysis approach which enables real-time generation of stable, comprehensive, and precise terrain models. Using successive LiDAR measurements through an NDT mapping technique, the method integrates spatial-temporal Bayesian generalized kernel inference and bilateral filtering, enhancing terrain elevation estimations in unobserved regions. The analysis of geometric connectivity between neighboring terrain cells facilitates traversability assessment and enables the differentiation of terrains with different traversal complexity. Experimental results using different datasets show the superiority of this method over existing approaches. Key contributions include the use of multiframe fusion, the introduction of bilateral filtering to reduce edge blurring, the inclusion of height variance as a weighting factor, and the generation of a travel costmap for path planning in complex environments.

ForestTrav, a novel method for autonomously assessing terrain traversability using LiDAR data in heavily vegetated areas proposed by Ruetz et al. [46] utilizes a 3D probabilistic voxel representation combined with Structural Contextual Networks (SCNNs) to continuously fuse LiDAR measurements and estimate per-voxel traversability. Evaluation using Matthews Correlation Coefficient[2] (MCC) demonstrates ForestTrav's superior performance over state-of-the-art methods, achieving MCC scores of 0.62 at 0.1 m voxel resolution, significantly outperforming alternatives (the closest competitor scored 0.41 at the same resolution) in discriminating between traversable and non-traversable terrain elements. The method proves robust across different vegetation densities and shows promise for real-time deployment in challenging, unstructured environments, highlighting its potential to improve autonomous navigation systems in natural settings.

[2] MCC is a measure used to evaluate the quality of binary classifications that is suitable for imbalanced data sets. It considers all four classification cases (true and false positives and negatives), while also offering invariance to class swaps.

4.2 Exteroceptive learning-based approaches

Learning-based approaches are becoming increasingly popular for traversability analysis, as new breakthroughs in neural network models are achieved, mostly using exteroceptive sensor modalities.

Guan et al. [6] presented a Terrain Traversability Mapping and Navigation System (TNS) for autonomous excavator applications in unstructured environments. Although it has potential, its performance has not been validated against state-of-the-art techniques such as [34]. By using RGB images and 3D point clouds, the system creates a global map for real-time planning and navigation. The approach adapts to changing environments and incorporates a novel learning-based geometric fusion solution that improves the accuracy of map generation and reduces errors in the traversability map. The system, deployed in real-world scenarios, claims a 49.3% improvement in overall success rates, navigating through complex terrains including deep pits and steep hills. In addition, the study made the Complex Worksite Terrain (CWT) dataset available.

Chen et al. [47] introduced ALTER, an adaptive off-road perception system that enables real-time prediction of traversability in unknown terrains. Using a self-supervised approach, the system dynamically adapts a visual model with short-range LiDAR data to improve the quality of predictions from visual features at longer ranges, showing an improvement in overall robustness in new environments. The adaptive visual method has been validated with off-road datasets and shows rapid terrain differentiation within one minute of data acquisition and training, providing fine-grained traversability. The study highlights the superior long-range accuracy of visual predictions. Figure 8 shows ALTER's capability of generating high-quality traversability estimations from visual data, as well as the fast adaptation of the model.

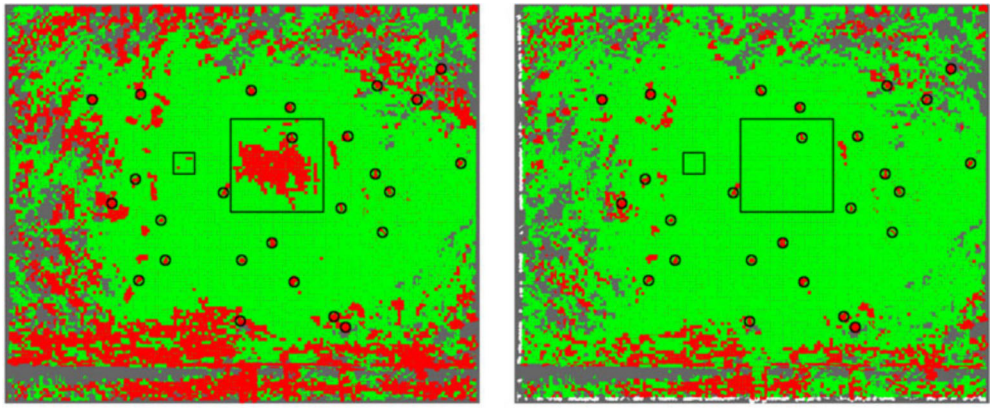


Fig. 7 Output of NDT-TM [36] with two different classifiers, showing the ability of the proposed one (on the right) to correctly characterize a group of tall weeds (the black rectangle in the center) as traversable as opposed to a rigid obstacle. Picture reproduced from the mentioned work

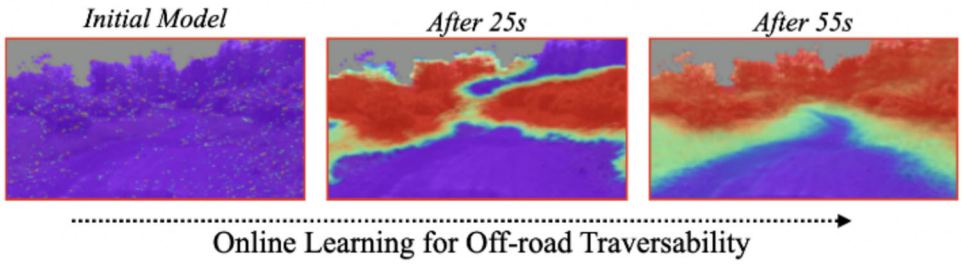


Fig. 8 Results of traversability analysis on RGB images over time using the approach proposed by Chen et al. [47]. Image from the original publication

Seo et al. [48] proposed a meta-learning framework for off-road traversability estimation. The traversability prediction network uses vehicle-terrain interactions to generate accurate continuous-valued costmaps from LiDAR data. Meta-learning is employed to train a global model, reducing uncertainty and enabling accurate predictions across diverse environments. Online adaptation further enhances predictions using recent interaction experiences. Experiments confirm the method's efficacy, ensuring stable off-road navigation in unstructured terrains. This framework holds promise for the deployment of autonomous robots in complex environments, with ongoing real-world evaluations and future plans to enhance it through sensor fusion, including LiDAR-RGB camera integration.

Guan et al. [49] presented GA-Nav, a novel group-wise attention mechanism designed to identify safe and navigable regions in unstructured environments using RGB images. It extracts multi-scale features from different terrain types independently and classifies them based on navigability using coarse-grained semantic segmentation. GA-Nav can be integrated into any backbone network, enhancing the focus on various terrain features while maintaining efficiency and high accuracy. Extensive evaluations on the RUGD and RELLIS-3D datasets show that GA-Nav achieves state-of-the-art performance, while real-world tests using Clearpath Jackal and Husky robots interfaced with a deep reinforcement learning-based navigation algorithm showed that GA-Nav improved navigation success rates and trajectory selections. The study highlights GA-Nav's stability across different terrains and its effectiveness in real-world scenarios when compared to existing methods. Future work will address labeling consistency and challenges posed by drastic elevation changes and extreme weather conditions.

4.3 Proprioceptive-based approaches

In recent years, significant work has been proposed that use diverse approaches and innovative data sources to estimate traversability, such as locomotion data from several UGVs combined with aerial data [50] and also proprioceptive feedback from legged [51] and wheeled robots [52]. Although various methods within this section are also learning based, they are specifically designed to work with proprioceptive data.

Zimmermann et al. [53] stand out by introducing Adaptive Traversability (AT) as a concept for autonomous motion control, adapting robot morphology to navigate complex terrain optimally. It was validated through a reinforcement learning based AT algorithm tested on natural obstacles, yielding positive results while reducing the cognitive load of the operator.

Haddeler et al. [54] presented a method that, based on human behavior, integrates terrain probing strategies into a traversability analysis framework to ensure safe navigation in rough, unstructured terrain. The framework supplements traditional vision and geometric information with collapsibility data, gathered through a force sensor, to assess the risk of terrain collapse. The method was validated on a quadruped robot, AlienGo, both in simulation and real-world experiments. The study demonstrates that combining vision and terrain probing significantly enhances the robot's ability to avoid hazardous regions. Future work includes enhancing the probing arm and exploring alternative probing methods integrated into the robot's gait.

Wellhausen et al. [55] presented a method for enhancing the navigation capabilities of legged robots on diverse terrains by predicting terrain properties ahead of the robot using a neural network. Data are collected during teleoperated experiments with a quadrupedal robot, where foothold positions are projected onto on-board camera images and labeled using force-torque signals from sensorized feet. The neural network is trained to generate dense predictions of terrain properties, which are used for both weakly supervised semantic segmentation and autonomous navigation. The approach is validated through extensive field experiments with the ANYmal quadruped robot, demonstrating high accuracy in terrain classification and effective path planning that avoids unsafe terrain. The results highlight the potential of this method to improve robot navigation by incorporating learned ground reaction properties and terrain classifications from visual data.

Sathyamoorthy et al. [52] introduced TerraPN, a self-supervised learning method for autonomous robot navigation over complex outdoor terrains. TerraPN uses RGB images of terrain surfaces and the robot's velocities as inputs, with IMU vibrations and odometry errors as labels, to generate a surface costmap that distinguishes between various terrain properties such as traction and bumpiness. This map helps the robot avoid difficult terrains by computing cost-based acceleration limits and dynamically feasible trajectories. Evaluated on a Clearpath Husky robot in diverse scenarios (including pavement, gardens, and forest), TerraPN demonstrates superior navigation performance with higher success rates and lower vibration costs compared to previous methods. The method also boasts faster training times and lower inference times due to efficient non-uniform patch sampling. Despite its effectiveness, TerraPN faces limitations with non-traversable surfaces, significant lighting changes, and subtle elevation differences, suggesting the need for future enhancements through multisensor fusion.

Gan et al. [56] proposed a novel approach to terrain traversability modeling for legged robots using Deep Inverse Reinforcement Learning (DIRL). The method addresses two key challenges: capturing the agile motions of legged robots, which are often missed by existing DIRL methods relying solely on environmental or kinematic features, and overcoming the suboptimal behaviors observed in robot demonstrations. By leveraging both exteroceptive environmental data and proprioceptive inertial features learned from robot-specific sensory inputs, the proposed deep neural network constructs a comprehensive reward function. This function not only enhances model accuracy but also refines policy decisions based on the robot's state during operation. Additionally, the study introduces the T-MEDIRL algorithm, integrating trajectory ranking based on locomotion energy consumption to improve policy efficiency beyond demonstration capabilities. Evaluation using a dataset from a Mini-Cheetah quadruped robot demonstrates the effectiveness of the approach, highlighting its potential for improving legged robot navigation in varied terrains.

Triest et al. [57] introduced an innovative approach to designing costmaps for off-road driving using inverse reinforcement learning (IRL). Addressing challenges in existing methods, which often suffer from over-confident predictions and lack of hardware validation, the authors propose a novel framework that integrates IRL with model-predictive control and risk estimation. They demonstrate significant improvements over geometric baseline methods, achieving a 44% improvement in reproducing expert driving paths and a 57% reduction in human interventions during practical tests. By varying the risk tolerance parameter (CVaR), the method exhibits adaptive navigation behaviors, crucially affecting performance in scenarios such as slopes and tall grass. Large-scale experiments in a forest environment validate these findings, highlighting the method's efficacy in real-world off-road navigation tasks. The study concludes with prospects for further enhancements, including the incorporation of visual features and velocity dimensions into the costmap design process.

Wang et al. [58] presented a framework for generating costmaps to assess terrain traversability for unmanned ground vehicles navigating off-road environments. The framework addresses several challenges, including the complexity and uncertainty inherent in off-road terrain. Using a combination of LiDAR-inertial odometry (LIO) and historical data, the proposed method generates local costmaps without requiring prior information. It incorporates factors such as slope, roughness, and potential slip risks to accurately describe terrain characteristics. To enhance stability, spatial constraints are applied to account for ego-motion uncertainty caused by UGV jolts. Experimental results across various test sites demonstrate the framework's effectiveness in generating stable costmaps with detailed traversability descriptions, facilitating safer and more efficient UGV navigation in challenging environments.

Vecchio et al. [59] addressed terrain traversability estimation for robot navigation over uneven surfaces, focusing



(a) Input synthetic RGB image.

(b) Terrain traversability estimation generated by the proposed system. The red overlay represents non-traversable regions.

(c) Input real RGB image.

(d) Terrain traversability estimation generated by the proposed system. The blue overlay represents non-traversable regions.

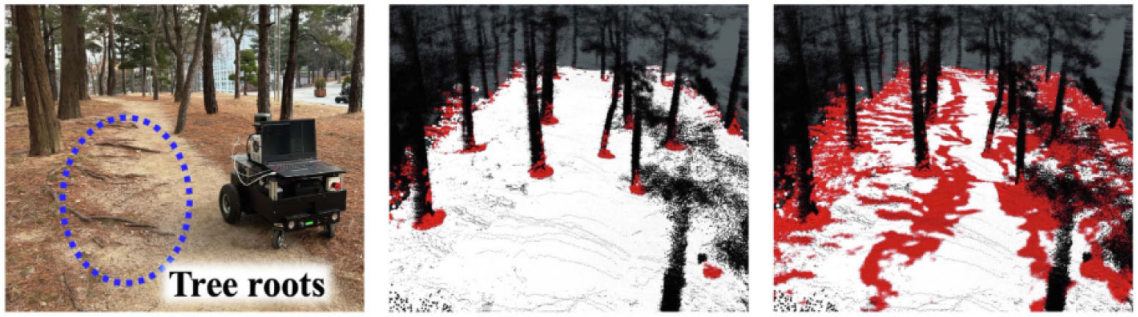
Fig. 9 Example result presented by Vecchio et al. [59] for the proposed approach. Picture reproduced from the original publication

on reducing the need for manual annotation in training datasets. The proposed method combines dataset synthesis, self-supervision, and unsupervised domain adaptation. It formulates traversability prediction as a vector regression task across vertical bands of observed frames, as seen in Fig. 9, leveraging pre-training via self-supervision to align synthetic and real data distributions. Supervised training on synthetic videos is enhanced with unsupervised domain adaptation to improve generalization to real-world scenarios. Experimental results demonstrate comparable performance to supervised approaches, supporting autonomous robot navigation without manual annotation requirements. The method's effectiveness is validated through on-field tests in challenging, unstructured environments, showcasing its potential for real-world deployment and future improvements aim at handling diverse terrain and environmental conditions.

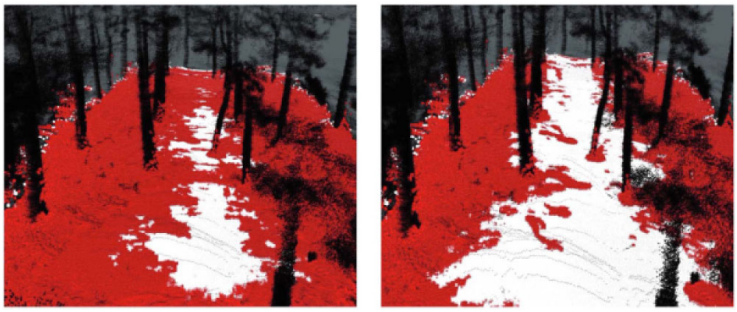
Gasparino et al. [60] proposed WayFASTER, a method for predicting traversability in complex outdoor environments using sequences of RGB and depth images fused with pose estimations. Employing a self-supervised neural network trained on receding horizon estimator data, WayFASTER predicts traversability without heuristic reliance, enhancing the robot's awareness of potentially navigable terrains like tall grass. Experimental validation demonstrates superior obstacle avoidance compared to baseline methods, especially in environments requiring wide field-of-view perception. Real-world tests on a TerraSentia 4-wheeled robot highlight WayFASTER's capability to navigate challenging paths involving tall grass and sharp turns, outperforming alternative approaches by maintaining visibility of traversability even when obstacles are temporarily occluded. The method's adaptability across platforms, including quadrupeds, underscores its robustness and potential for future enhancements integrating real-time network model adaptability.

Elnoor et al. [61] introduced AMCO, a novel navigation method tailored for quadruped robots navigating complex outdoor environments by dynamically combining vision-based and proprioception-based perceptions. The approach integrates three distinct costmaps: a general knowledge map derived from semantic segmentation of visual data, a traversability history map based on recent proprioceptive measurements, and a current proprioception map reflecting real-time sensor data. By adaptively coupling these maps, AMCO optimizes traversability predictions, enabling effective gait and velocity selection for navigating challenging terrains such as sand, dense vegetation, and mud. Experimental validation on a Boston Dynamics Spot robot shows AMCO's significant performance improvements over existing methods, achieving up to 50% higher success rates and reducing stability metrics by 10.8% to 34.9% across various test scenarios. One notable limitation is AMCO's reliance on proprioceptive data under dark conditions, which restricts its effectiveness in environments with low light levels.

Cho et al. [62] presented a novel self-supervised traversability learning method designed for wheeled mobile robots navigating diverse outdoor terrains. The approach addresses the challenge of providing robots with sufficient and varied experience by focusing on high-risk terrains using a risk-sensitive instance weighting scheme. Leveraging a neural network trained with confidence-regularized self-training,



(a) On the left pane is a representation of the experimental environment. On the center and right, the method from Shan et al. [35] with two different parametrizations.



(b) On the left, a self-training method by Lee et al. [33] used for comparisons, and on the right the proposed method, where it is visible the improvement in terrain segmentation quality.

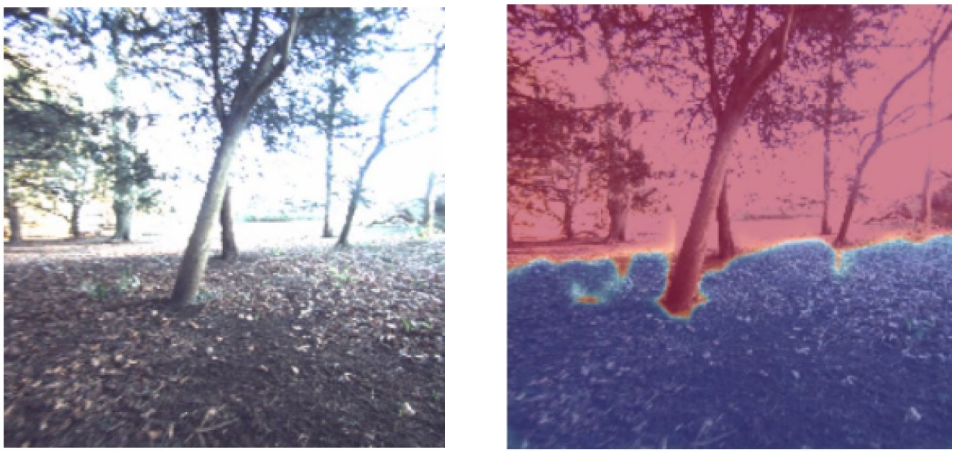
Fig. 10 Comparative results presented by Cho et al. [62]. Figures adapted from the original publication

the method generates high-quality pseudo-labels for unexperienced terrains, thereby augmenting the training dataset effectively. Extensive real-world experiments in both urban and agricultural environments demonstrate the method's efficacy in accurate traversability estimation across varied terrains, including asphalt, cobblestones, gravel paths, and grassy areas with obstacles like curbs and tree roots. Comparative evaluations against conventional methods (such as the one based in BGK presented in [35]) highlight superior performance (see Fig. 10), especially in challenging terrains, underscoring its potential for enhancing safe and efficient robot navigation in complex outdoor settings.

Yoon et al. [63] introduced the ARTE framework with Experience Replay with Uncertainty (ERU) to tackle challenges in self-supervised learning for robot navigation across unstructured environments. Current self-supervised methods often fail to adapt to changing environments and neglect dataset uncertainty critical in such contexts. ARTE leverages online continual learning to adapt to real-time conditions and ERU integrates uncertainty-aware training to enhance traversability estimation robustness. Real-world experiments with Clearpath's Husky robot across varied environments demonstrate ARTE's superior navigation performance over offline and fine-tuning baselines, notably handling vegetation-rich terrains. Evaluation metrics show ARTE's effectiveness, particularly in reducing traversal discomfort metrics across all tested environments. Additionally, benchmark evaluations against ImageNet and CORe50 datasets affirm ERU's efficacy in managing dataset uncertainties crucial for self-supervised applications in dynamic environments.

Mattamala et al. [64] presented an open-source implementation of a self-supervised learning approach based on vision and proprioceptive feedback. The method automatically extracts features from a visual transformer and matches them with a corresponding proprioceptive feedback, enabling it to gradually build confidence about certain features from the robot's experience. It works in real-time, performing online adaptation to new terrains when encountered. Figure 11 represents the output of their technique.

Atas et al. [65] introduced a deep learning-based approach for terrain traversability estimation, addressing challenges in autonomous navigation in outdoor environments like forests. The proposed neural network processes raw LiDAR and IMU data to generate continuous traversability assessments using spatiotemporal point cloud maps. By leveraging a high-fidelity simulation pipeline for automated data generation and labeling, the method eliminates the need for extensive real-world data collection, enabling efficient large-scale training. Experimental validation on both simulated and real-



(a) Input RGB image.

(b) Output of the WVN system generating a pixel-wise terrain traversability score (colder colors represent traversable terrain).

Fig. 11 Example result presented by Mattamala et al. [64] for their Wild Visual Navigation (WVN) approach. Picture adapted from the original publication

world platforms highlights the method's superior predictive accuracy and adaptability, even without prior exposure to real-world data. Furthermore, the traversability predictions are integrated into 3D path planning algorithms, consistently generating collision-free paths and demonstrating practical effectiveness in real-world scenarios. To encourage further research and application, the implementation has been made publicly available, underscoring the approach's utility and potential for advancing autonomous navigation technologies.

4.4 Comparative analysis and research directions

In this section, we present a table with a comparison between all methods, as well as some remarks and trends identified in said comparison.

Tables 1 and 2 present the qualitative comparison between the methods described. The focus of this table was on methods developed for dealing with unstructured environments such as forests. The majority of the works therein are developed for the assessment of unstructured terrain and tested in these very scenarios, using appropriate metrics. Namely:

- Dynamic Obstacles—Whether a method considers the possibility of encountering dynamic obstacles and integrates these obstacles explicitly into its behavior;
- Environment—Whether the proposed method is designed to deal with structured or unstructured environments;
- Urban or Rural—Helps to differentiate methods according to the type of sub-environment for which they were developed (unstructured urban, unstructured rural, ...);
- Tested in Forest—This metric further refines the categorization of methods according to the specificity of their target terrain, because although all forests are (at least partially) unstructured, not all unstructured environments are necessarily forests (e.g., earthquake disaster scenarios);
- Dataset or Sim. Data—Distinguishes work in which the authors used datasets comprised of real-world data from works that only used simulated data in their experiments;
- Real World Tests—Whether the technique was tested with a real robotic platform (if applicable) and real sensors in real-world conditions;
- Sensors Used—The type of sensors the method uses to produce the presented results;
- Online—Whether or not a method is executed in real time. In the context of this article, "real time" means that a method is capable of generating traversability estimates at such a rate that it does not present a bottleneck to the rest of the system during operation;
- Map Type—The type of traversability representation used by the proposed method (typically 2D or 3D);
- Ground Truth—Whether the authors have access to a ground-truth traversability estimation and use it to evaluate the performance of the proposed technique.

Tables 1 and 2 list the most relevant recent works on traversability analysis in unstructured environments and compare them using reference metrics.

Most noticeable, the overwhelming majority of recent work relies—at least partially—on 3D LiDAR sensors to perceive the environment. This result is not surprising, given how much these sensors have evolved in recent years, achieving unprecedented levels of precision and performance at a fraction of the price [68]. However, LiDAR sensors—and optical sensors in general—struggle in environments with a large amount of airborne particles such as dust and fog, which is why sensor fusion with non-optical sensors (such as Ultra-Wideband RADAR) has been explored with some success in the past [69].

Table 1 Qualitative comparison between traversability analysis methods

Works	Paradigm	Dynamic obstacles	Environment	Urban or rural	Tested in forest	Dataset or Sim. Data	Real-world tests	Sensors used	Online	Map type	Ground truth
Mattamala et al. [64]	Vision and proprioception	✓	Both	Both	✓	Dataset	✓	RGB Camera + Velocity Sensor	✓	3D	✗
Yoon et al. [63]	Extero and proprioceptive self-supervised learning	✗	Unstructured	Rural	✓	Dataset	✓	3D LiDAR + IMU	✓	2D	✗
Cho et al. [62]	Risk-aware, extero and proprioceptive self-supervised learning	✗	Both	Both	✓	Dataset	✓	3D LiDAR	✓	3D	✓
Elnoor et al. [61]	Vision and proprioception	✗	Both	Both	✗	None	✓	3D LiDAR + RGB Camera + Joint Feedback + Current Sensor	✓	2D	✗
Gasparino et al. [60]	Extero and proprioceptive self-supervised learning	✗	Unstructured	Rural	✓	Dataset	✓	RGB-D Camera	✓	2D	✗
Elnoor et al. [51]	Proprioceptive traversability estimation	✗	Unstructured	Both	✓	None	✓	Joint Feedback + Current Sensor	✓	N/A	✗
Gan et al. [56]	Deep inverse reinforcement learning	✗	Unstructured	Both	✓	Both	✓	RGB-D Camera + IMU	✗	2D	✗
Sathyamoorthy et al. [52]	Extero and proprioceptive self-supervised learning	✗	Unstructured	Both	✓	Dataset	✓	RGB Camera + Velocity Sensor	✓	2D	✗
Wellhausen et al. [55]	Extero and proprioceptive self-supervised learning	✗	Unstructured	Both	✗	Dataset	✓	RGB Camera + Foot Force Sensors	✓	2D	✗
Ruetz et al. [46]	LiDAR-only probabilistic approach	✗	Unstructured	Rural	✓	Dataset	✓	3D LiDAR	✓	3D	✓
Vecchio et al. [59]	Self-supervised learning + Unsupervised domain adaptation	✗	Unstructured	Rural	✓	Both	✓	RGB Camera	✓	3D	✗
Triest et al. [57]	Deep inverse reinforcement learning	✗	Unstructured	Rural	✓	Dataset	✓	RGB-D Camera + LiDAR	✓	2D	✗
Haddeler et al. [54]	Vision + Terrain probing	✗	Unstructured	Both	✗	Simulated	✓	RGB-D Camera + Force Probe	✓	3D	✗
Wang et al. [58]	Self-supervised learning	✗	Unstructured	Rural	✓	Dataset	✓	3D LiDAR + IMU	✓	3D	✗
Castro et al. [66]	Self-supervised learning	✗	Unstructured	Rural	✓	Dataset	✓	RGB Camera	✓	3D	✗
Guan et al. [49]	Group-wise attention mechanism	✗	Both	Both	✓	Dataset	✓	3D LiDAR + RGB Camera	✓	3D	✓
Carvalho et al. [67]	Mechanical effort	✗	Unstructured	Rural	✓	Simulated	✗	3D LiDAR	✗	3D	✗
Seo et al. [48]	Self-supervised meta-learning	✗	Unstructured	Both	✓	Both	✓	3D LiDAR + IMU	✓	2D	✗
Chen et al. [47]	Self-supervised learning	✗	Unstructured	Rural	✓	Dataset	✗	3D LiDAR + RGB Camera	✓	3D	✗

Table 2 Qualitative comparison between traversability analysis methods—continuation

Works	Paradigm	Dynamic obstacles	Environment	Urban or rural	Tested in forest	Dataset or Sim. Data	Real-world tests	Sensors used	Online	Map type	Ground truth
Guan et al. [6]	Geometric + Semantic Analysis	✗	Unstructured	Rural	✗	Dataset	✓	3D LiDAR + RGB Camera	✓	3D	✓
Xue et al. [45]	NDT Information Fusion	✓	Unstructured	Both	✗	Dataset	✓	3D LiDAR	✓	3D	✓
Waibel et al. [25]	Statistical Approach and 2D Neural Network	✗	Unstructured	Both	✗	None	✓	3D LiDAR + IMU	✓	2D	✓
Ruetz et al. [34]	Forest Traversability Map	✗	Unstructured	Rural	✓	None	✓	3D LiDAR + RGB Camera	✓	3D	✗
Wallin et al. [32]	DNN with HeightMaps	✗	Unstructured	Rural	✗	Both	✗	3D LiDAR	N/A	3D	✓
Lee and Chung [33]	Self-Training NN Approach	✓	Structured	Urban	✗	Dataset	✗	3D LiDAR	✗	3D	✗
Wang et al. [43]	Multi-Layer 2D maps from OctoMap	✗	Structured	Urban	✗	Simulated	✓	2D LiDAR + RGB-D Camera	N/A	3D	✗
Shan et al. [35]	Bayesian Generalized Kernel Inference	✗	Both	Both	✗	Both	✓	3D LiDAR	✗	3D	✓
Chavez-Garcia et al. [30]	CNN	✗	Both	Both	✗	Both	✓	RGB-D Camera	✓	3D	✗
Yang et al. [37]	2.5D-NDT	✗	Both	Urban	✗	Both	✗	3D LiDAR	✗	2.5D	✗
Krusi et al. [40]	Computations directly in Unordered Point Clouds	✓	Both	Both	✓	None	✓	3D LiDAR + IMU + Wheel Encoders	✓	3D	✗
Ahtiainen et al. [36]	NDT-TM	✓	Both	Both	✓	None	✓	3D LiDAR	✓	3D	✓
Putz et al. [38]	3D Navigation Mesh	✗	Both	Both	✓	None	✓	3D LiDAR	✓	3D	✗
Tai et al. [31]	Neural Network Depth Map	✗	Structured	Urban	✗	Simulated	✓	RGB-D Camera	✓	3D	✗
Zimmermann et al. [53]	Reinforcement Learning Adaptive Traversability	✗	Unstructured	Both	✓	Dataset	✓	3D LiDAR + Omni-Directional Camera	N/A	3D	✗

4.5 Common limitations of current techniques

Although most of the aforementioned works focus on unstructured environments, many of them do not explicitly deal with forests. Although this type of environment is one of the most challenging and least studied, it has recently increased in popularity significantly, with state-of-the-art methods such as NDT-TM [36], FTM [34], and WVN [64] being proposed. However, there are some challenges in forest robotics that have not yet been solved in the literature, such as:

- The stability and robustness of the proposed methods which has to be achieved in order to cope with all complex non-structured environments;
- Real-time processing in forest robotics is computationally expensive due to the high data rates from multiple sensor modalities (e.g., LiDAR, cameras, and IMUs). This often limits their joint deployment on mobile robotic platforms with constrained processing power, necessitating the development of lightweight algorithms or hardware-accelerated solutions;
- The visual likeliness of most forest environments, which makes it hard to use pure visual-based techniques to navigate them;
- Depending on how dense the forest is, GNSS signal may be partially or completely blocked by obstacles such as tree canopies;
- Recognizing and traversing extremely rough or dangerous terrain autonomously. Current methods often fail in environments with obstacles such as fallen trees, large rocks, or swampy areas, which require both advanced perception and decision-making capabilities;
- Some sensor modalities such as RADAR, multispectral and thermal imaging offer interesting nuance in some scenarios, as discussed above (vegetated areas, low visibility conditions, etc.), but their price point, size, and power requirements make them hard to integrate in modern techniques;
- Ensuring that the environment in general, but fauna and flora in forests in particular, remain as shielded as possible from our activities while still achieving our goals.

To address these limitations, future research could focus on developing more computationally efficient algorithms, integrating multisensor fusion techniques for robust perception, and creating methods that adapt to the dynamic and variable nature of forest environments. Highlighting failure cases and addressing these gaps—such as enhancing GNSS-free localization techniques or improving computational efficiency for sensor-heavy systems—could lead to more reliable practical deployments in forest robotics.

4.6 Research trends

From the analysis of this section and the comparison presented in Tables 1 and 2, we can identify some latent trends. First, the use of 3D methods to represent the environment is widespread, and the ability to do this in real time is explainable given the major breakthroughs in computing power and storage capacity over the last decade. Furthermore, an increasing number of works [34, 36, 45] propose the use of NDT as the backbone of their approaches and have been successful in doing so, even in 2.5D representation paradigms [37]. There are also various recent techniques using sparse kernels in their architectures [35, 45, 62, 70], which allow, among other things, for fast inference in sparse LiDAR datasets. This drastically improves mapping performance with sparse LiDAR data and enables real-time performance in low power processing units, for instance.

The use of proprioceptive feedback is also gaining popularity, as researchers find that it can be used effectively to make neural network models learn what to expect from exteroceptive data [52, 61, 62, 64]. Furthermore, proprioceptive data are more nuanced and can provide higher quality information regarding the texture and stability of terrain that is visually challenging to differentiate through the robot's own experience. When combined with machine learning and exteroceptive sensor modalities [60, 61, 63, 64], proprioceptive feedback about the current and past experience of the robot can successfully be used to learn how to predict similar conditions in the future in terrains where only exteroceptive methods struggle. This approach has also been used in the past to estimate terrain properties, as presented in [71], and is once again gaining popularity, as shown above.

Finally, the use of machine learning is the highest ever in the context of traversability analysis, as new models for various tasks are proposed and successfully applied in the segmentation of images, point clouds, and other tasks. More specifically, methods that do not require direct human supervision (such as reinforcement learning and self-supervised learning) are used the most, as they mitigate the need for laborious hours of manual data annotation while yielding increasingly positive results.

5 Expectations and roadmap for the future

Undeniable advances have been made in recent years, such as the admirable increase in processing power with recent CPUs and GPUs that was mentioned in the previous section. Having this in mind, most of the emerging methods will likely rely on 3D representations of the environment or even direct processing of measurements from 3D range sensors such as LiDAR [40]. To achieve the 3D representations, LiDARs have been used in most recent works, a trend that is

expected to continue due to the aforementioned performance improvement and price reduction across the industry.

Both NDT and Bayesian Generalized Kernels (BGK), as discussed, have been showing promising results, which leads us to believe that future methodologies should consider integrating such algorithms in their architectures.

Regarding proprioceptive data and its recent success, as explored above, the use of quadruped robots is also expected to keep increasing in forestry tasks that allow it, as they provide rich proprioceptive feedback which is valuable for the modern learning architectures.

As a tentative roadmap for the future, we have identified the following points:

- Design future methods so that integration with various types of robotic platforms is as straightforward as possible.
- Further explore learning-based approaches, as these are delivering increasingly promising results.
  - Self-supervised methods in particular are becoming popular, as they partially or completely eliminate the need for hand-labeled datasets, which are extremely labor-intensive and time-consuming to create. They achieve similar (and in some cases better) results than their more traditional supervised counterparts.
- Real-Time Implementation, Dense Vegetation and Sensor Modalities:
  - Develop and refine real-time processing algorithms to instantly respond to dynamic environments.
  - Overcome the challenges posed by dense vegetation through advanced sensing and data processing techniques, enhancing the robot's ability to understand the characteristics of the observed scene such as its rigidity, thus achieving more natural and robust forest navigation;
  - Explore cheaper, lighter and overall less-demanding RADAR sensors to enable its integration in a larger variety of mobile platforms.
  - Explore newer sensor modalities, such as Frequency Modulated Continuous Wave (FMCW) LiDAR, which generates high-quality depth estimates even in environments with poor visibility (for example, fog and rain).
- Negative Obstacles and Computational Efficiency:
  - Focus on improving the detection of negative obstacles (e.g., holes, ditches) which are critical for safe navigation and represent one of the open challenges for proprioceptive-based self-supervised approaches, given the risk associated with acquiring robotic experience in these scenarios.
  - Further explore multi-resolution representations for balancing computational efficiency and detection completeness, enabling faster and reliable traversability analysis.
- Safety and Compliance with Machine Specifications:
  - Prioritize safety in the development of algorithms and ensure that autonomous systems can operate without risk to humans, animals or the environment.
  - In certain applications, adapting the traversability analysis algorithms to the specifications of the equipment (e.g., excavators) is an important step toward overall optimization across various machines.
- Online Learning and LiDAR Feature Integration:
  - Advance the integration of online learning mechanisms into visual-LiDAR models to actively adapt to new terrain and obstacles.
  - Integrate additional point cloud features such as geometry and LiDAR intensity into analysis models to gain a more comprehensive understanding of the terrain.
- Ethical and Sustainability Considerations:
  - Autonomous machines should use electrical power sources as much as possible in order to minimize their impact in the environment (such as pollution and noise);
  - The decision-making process of autonomous operators in forest environments should be transparent and carefully analyzed considering their potential impact in the ecosystem.
  - In contrast to traversability systems, semantic segmentation methods commonly assign a semantic class to people and animals. As such, combining this information with modern traversability analysis systems could generate more robust and safe systems overall.
- Collaboration and Autonomy:
  - Explore ways to enable more effective collaboration between machines and humans, leveraging each party's strengths for improved navigational decision-making in exceptionally complex or dangerous terrain.
  - Push the boundaries of autonomy in mobile robots, aiming for systems that can independently assess and adapt to unknown environments, while also being able to understand when minimal human intervention is required.
  - Improve collaboration between domestic and international research groups by publishing more open-

source projects, datasets and computational efficiency metrics.

6 Conclusion

This short survey examines the field of traversability analysis, from the classic methodologies to the latest breakthroughs, with a special focus on forestry applications. Approaches based on computer vision, LiDAR, multi-modal sensor fusion, and advanced learning techniques are presented and compared.

As discussed, challenges persist. The need for robustness, real-time adaptability, and the ability to handle unforeseen obstacles remains a focal point for future research. The integration of artificial intelligence in conjunction with sensor fusion opens up exciting avenues for exploration. Future efforts must focus on refining existing methods, exploring new research directions and interweaving the complexity of natural environments with the design of the methods themselves.

The unique challenges that forest environments present, such as dense vegetation, variable terrain, and dynamic obstacles, emphasize the importance of specific traversability analysis. Addressing these forest-specific challenges requires dedicated research efforts to develop more adaptable, accurate, and resilient navigation solutions. This highlights the need for continued research and innovation in traversability analysis specifically tailored to the complexity of forest landscapes, to ensure that autonomous systems can navigate safely and effectively in this challenging environment.

In conclusion, this review demonstrates the progress made to date, offering a panoramic view of traversability analysis in forest terrains.

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Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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